



UTS Communication
Protocol

Program Studi Sains
Data

Kelompok 05

Implementasi Communication Protocols Menggunakan HTTPS, Wireshark, XML Parsing, dan Python

Kelompok 05:

Aryadi

Faza Qinthoro Putra Tsany

Uswatun Hasanah

Nabiel Azzacky



Latar Belakang

Komunikasi data merupakan fondasi sistem digital modern. Proyek ini meliputi pengujian HTTPS dengan Postman, analisis paket jaringan menggunakan Wireshark, serta XML parsing dengan Python untuk menghasilkan data CSV yang terstruktur.



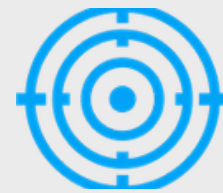


Rumusan Masalah



HTTPS GET & POST

Bagaimana mengimplementasikan metode HTTPS GET dan POST?



Analisis Paket Jaringan

Bagaimana menganalisis paket jaringan dengan Wireshark?



Pemrosesan Data XML

Bagaimana memproses data XML menjadi format terstruktur?

Tujuan Proyek

Melakukan parsing XML menggunakan Python dengan library ElementTree dan Pandas, serta menghasilkan output data terstruktur dalam format CSV yang siap dianalisis.

Komunikasi Data



Memahami konsep dasar komunikasi data dalam sistem digital modern

Implementasi HTTPS



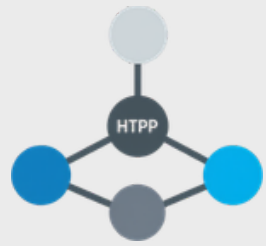
Menerapkan metode HTTPS GET dan POST untuk pertukaran data

Analisis Jaringan



Melakukan analisis paket jaringan menggunakan Wireshark

Dasar Teori



Communication Protocol

Aturan yang mengatur pertukaran data antar perangkat dalam jaringan. HTTPS digunakan sebagai protokol komunikasi utama.



Tools & Format Data

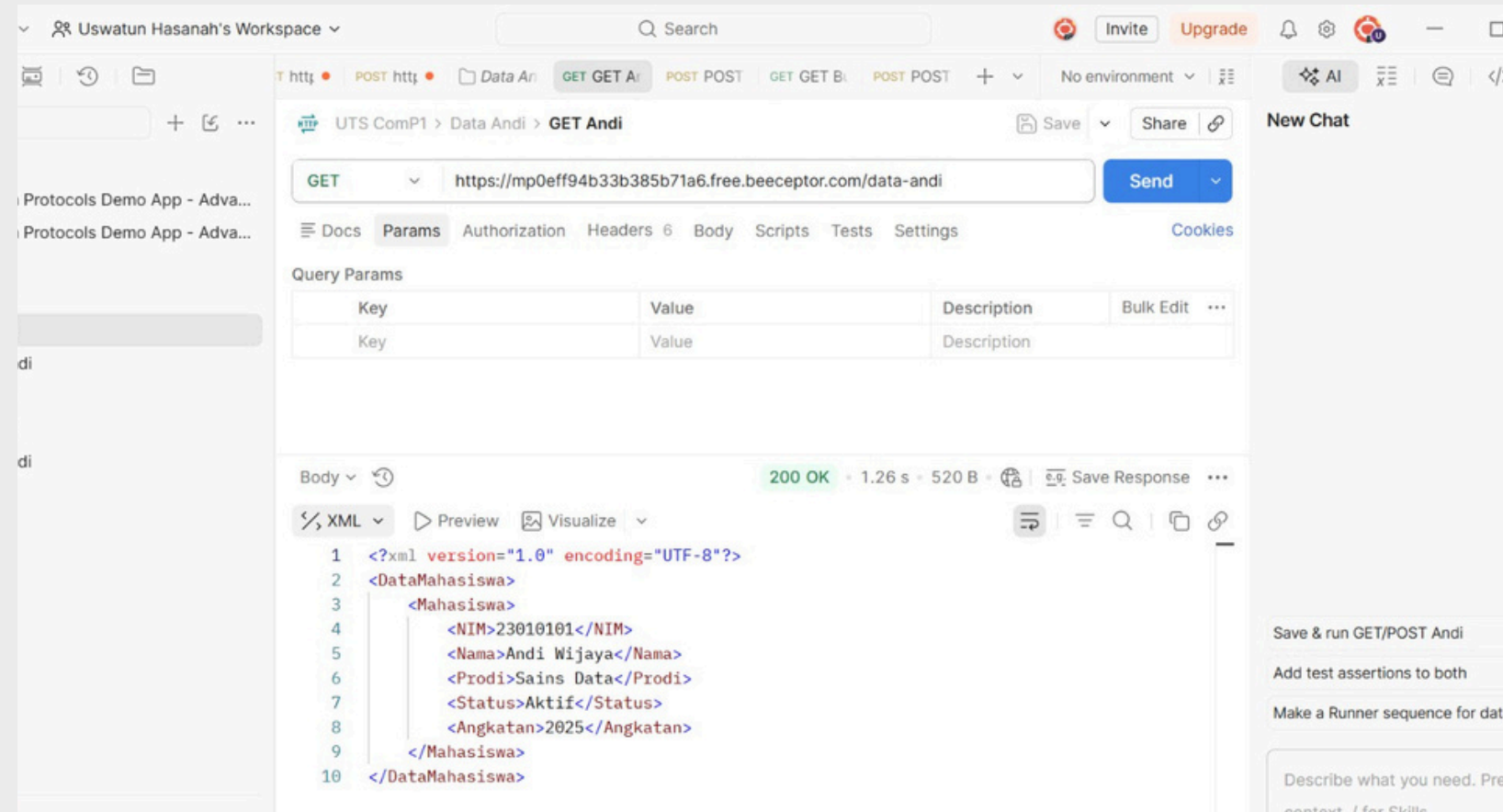
Wireshark untuk analisis jaringan, XML sebagai format data terstruktur, dan Python sebagai alat pemrosesan data.

Teknologi yang Digunakan



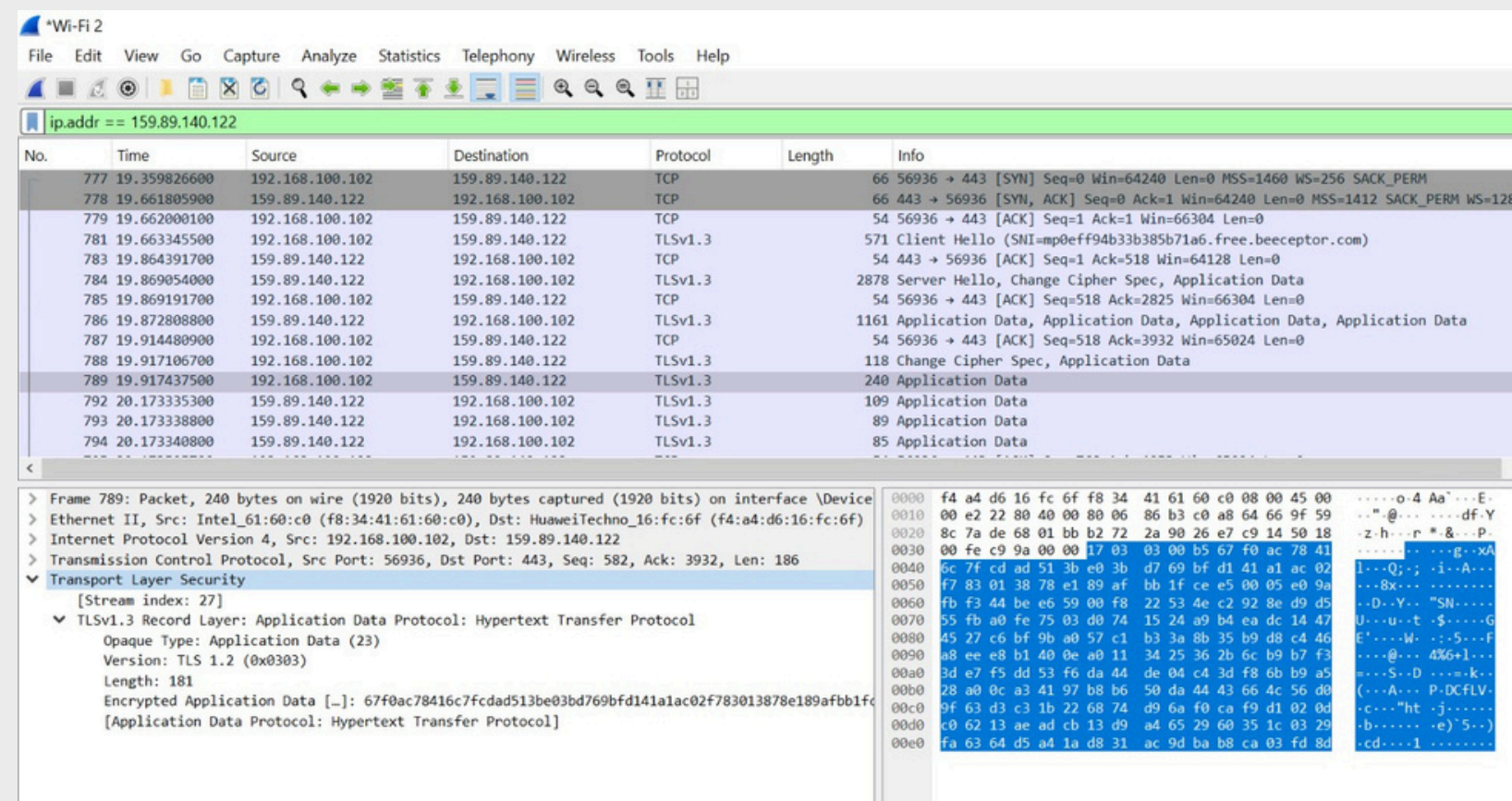
Postman & HTTPS

API client untuk pengujian
HTTPS GET dan POST ke
server



Wireshark

Analisis paket jaringan dan
monitoring komunikasi
TCP/HTTP



Python & Pandas

Parsing XML dengan
ElementTree, manipulasi
data, ekspor CSV



Diagram Arsitektur Sistem

Setiap komponen dalam arsitektur sistem memiliki peran spesifik dalam proses komunikasi dan transformasi data dari request hingga output akhir.



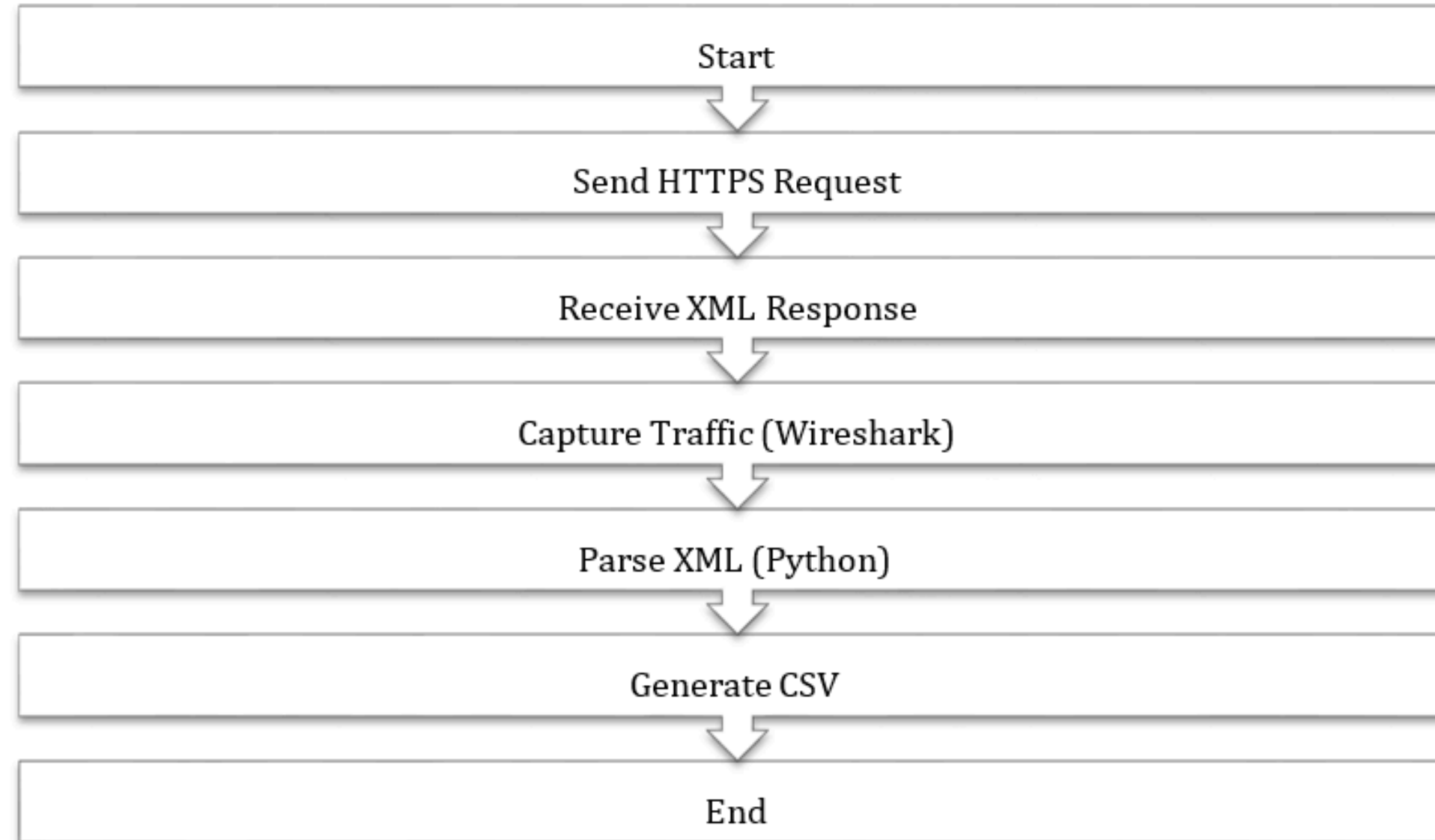
Alur data dari Client hingga output CSV melalui HTTPS, XML, dan Python secara terstruktur.



Client (Postman) mengirim HTTPS Request → Server/API memproses → XML Response diterima → Python Parser mengolah → CSV Output dihasilkan.

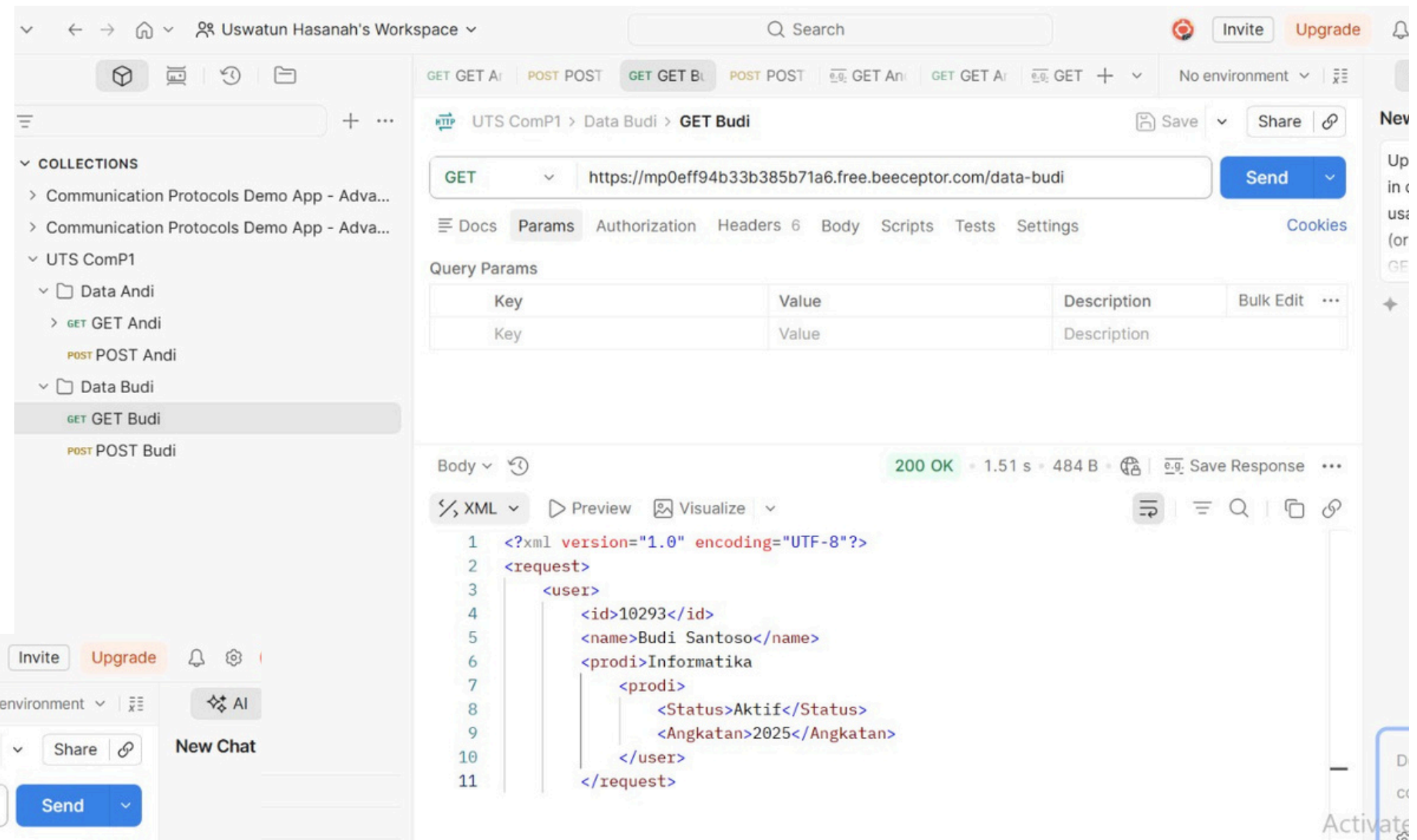
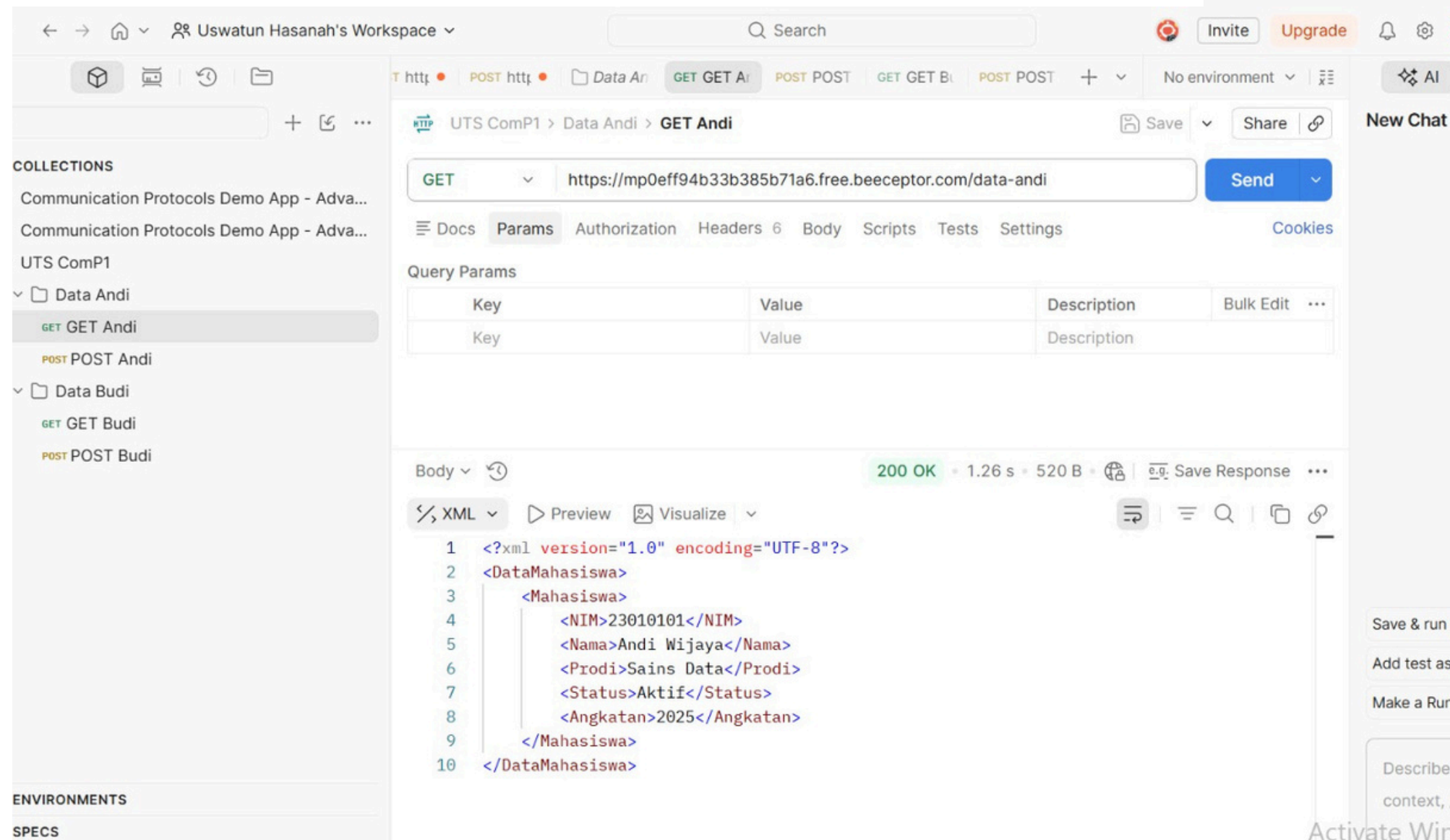


Flowchart System



Implementasi HTTPS GET

Metode GET digunakan untuk mengambil data dari server. Pengujian dilakukan menggunakan Postman dengan endpoint yang telah disediakan. Server merespons dengan data XML yang berisi informasi pengguna.



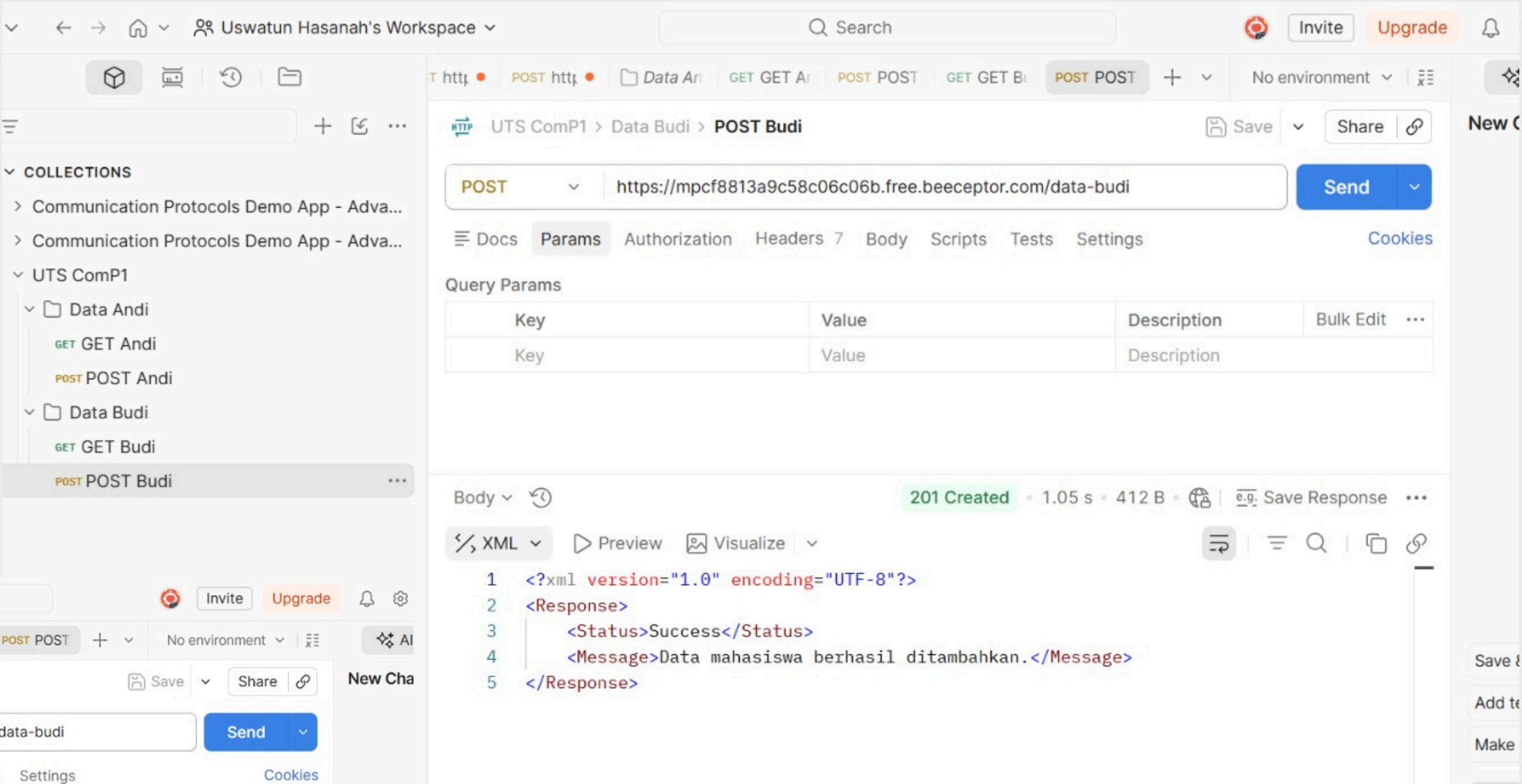
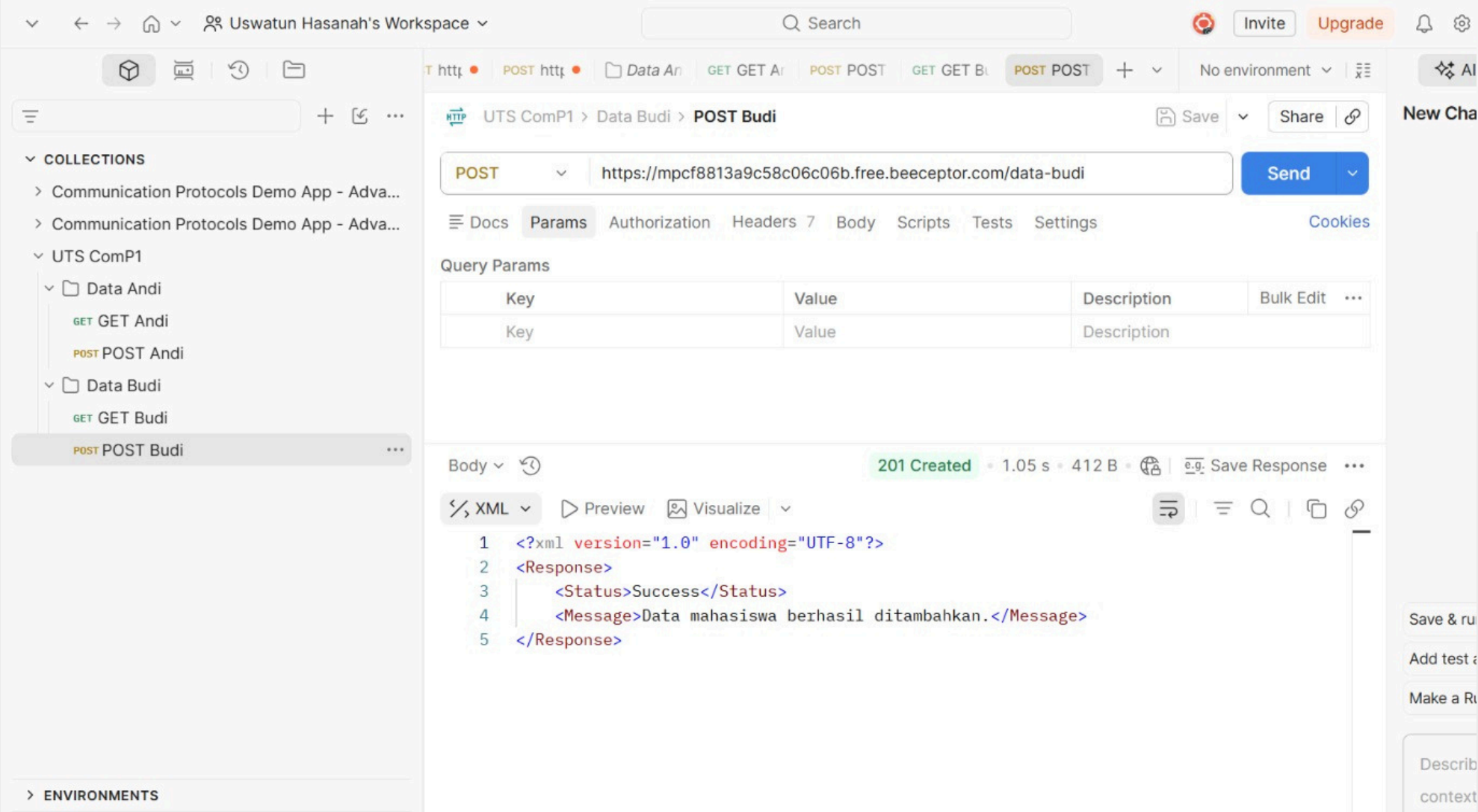
01 GET Data Budi

Status Code: 200 OK – XML response berhasil diterima dari server

02 GET Data Andi

Status Code: 200 OK – Data Andi berhasil diambil dalam format XML terstruktur.

Implementasi HTTPS POST



01 POST Data Budi

Status Code: 201 - Data berhasil dikirim dan diproses oleh server

02 POST Data Andi

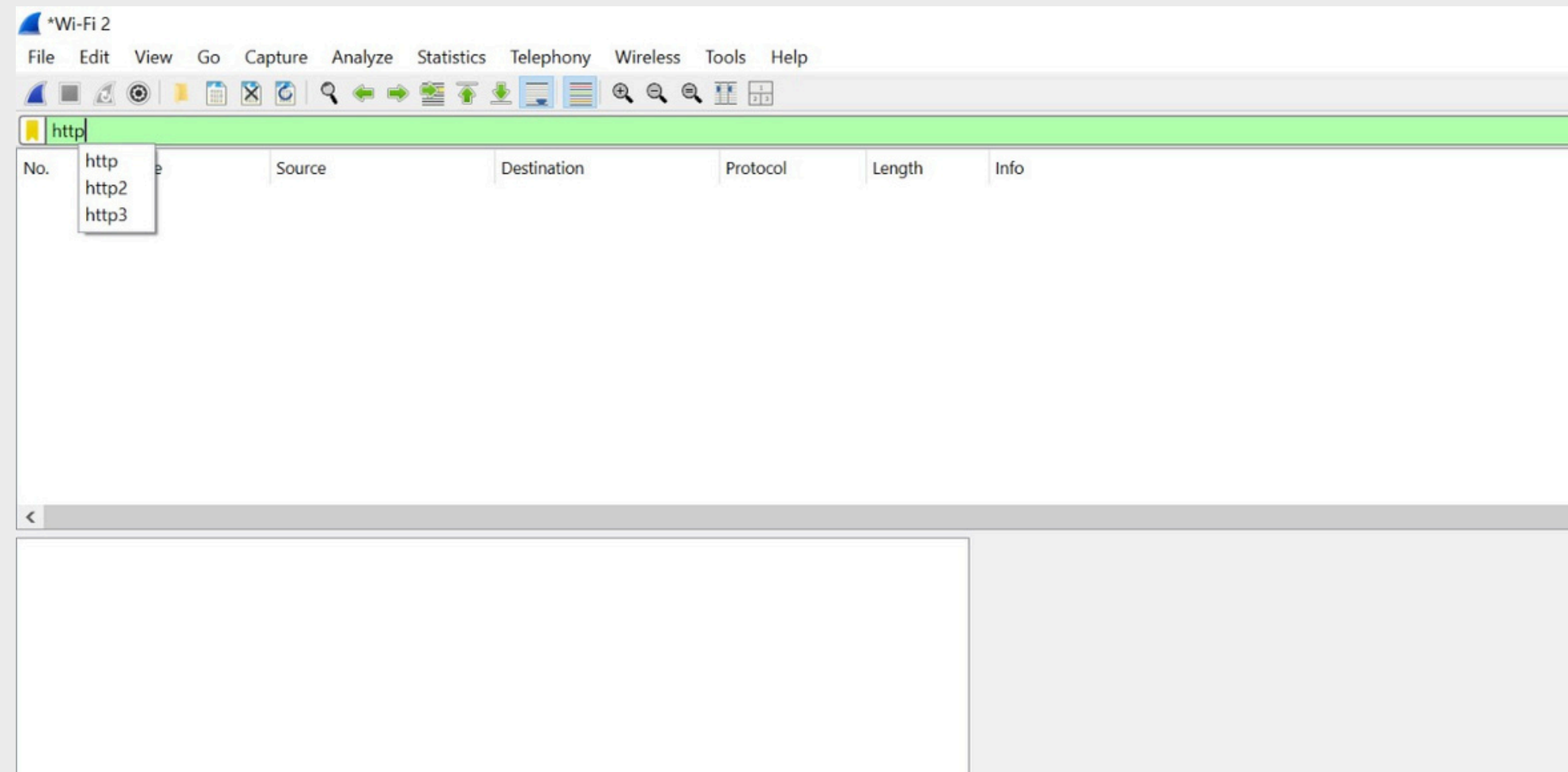
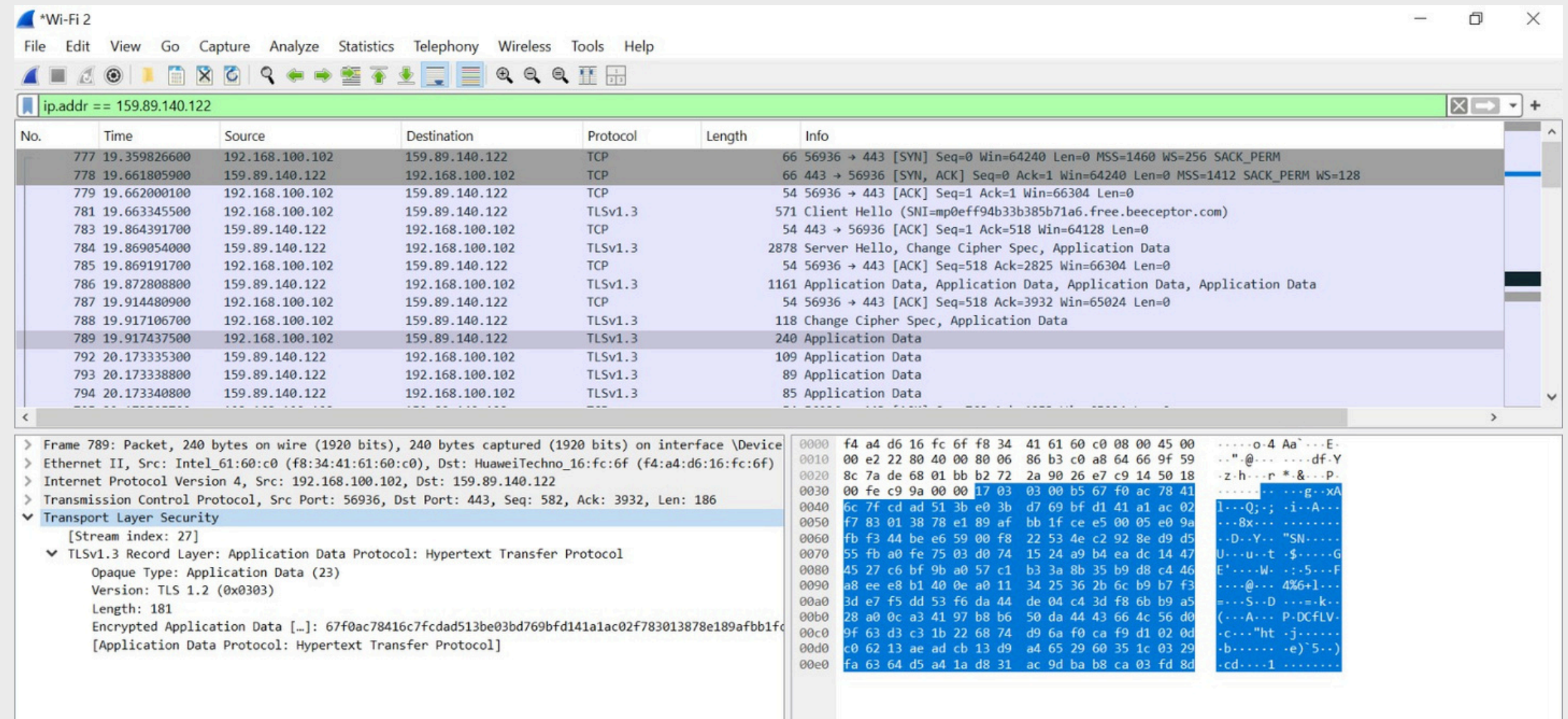
Status Code: 201 - Respon server mengonfirmasi data diterima dengan sukses

Analisis Wireshark

Hasil analisis traffic jaringan menggunakan Wireshark:

- TCP packets berhasil terkirim
- Komunikasi TLSv1.3 terdeteksi
- DNS Query dan Response tercapture
- HTTPS Request dan Response berhasil direkam

Sedangkan apabila kita filter dengan HTTP itu maka kosong/ tidak ada datanya. Karena kita pakai HTTPS yang notabennya ada perlindungan protokol TLS yang membungkus paket tersebut.



*Wi-Fi 2

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tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
91	3.957682700	2001:448a:2083:6c40:b9...	2606:4700::6810:ba82	TCP	86	56919 → 443 [SYN] Seq=0 Win=64952 Len=0 MSS=1412 WS=256 SACK_PERM
98	3.976411900	2606:4700::6810:ba82	2001:448a:2083:6c40:b9...	TCP	86	443 → 56919 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192
108	4.022582200	192.168.100.102	18.64.18.8	TCP	66	56920 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
109	4.022884000	192.168.100.102	18.64.18.8	TCP	66	56921 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
114	4.030379700	18.64.18.8	192.168.100.102	TCP	66	443 → 56921 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
116	4.030539500	18.64.18.8	192.168.100.102	TCP	66	443 → 56920 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
153	4.194015400	192.168.100.102	18.64.18.8	TCP	66	56922 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
175	4.211949200	18.64.18.8	192.168.100.102	TCP	66	443 → 56922 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
198	4.245194000	192.168.100.102	18.67.175.96	TCP	66	56923 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
199	4.251250200	18.67.175.96	192.168.100.102	TCP	66	443 → 56923 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
248	4.574847400	2001:448a:2083:6c40:b9...	2606:4700:3030::ac43:b...	TCP	86	56924 → 443 [SYN] Seq=0 Win=64952 Len=0 MSS=1412 WS=256 SACK_PERM
249	4.593895800	2606:4700:3030::ac43:b...	2001:448a:2083:6c40:b9...	TCP	86	443 → 56924 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192
259	4.619719600	2001:448a:2083:6c40:b9...	2606:4700:3030::ac43:b...	TCP	86	56925 → 443 [SYN] Seq=0 Win=64952 Len=0 MSS=1412 WS=256 SACK_PERM
261	4.641451900	2606:4700:3030::ac43:b...	2001:448a:2083:6c40:b9...	TCP	86	443 → 56925 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192

> Frame 175: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...
> Ethernet II, Src: HuaweiTechno_16:fc:6f (f4:a4:d6:16:fc:6f), Dst: Intel_61:60:c0 (f8:34:41:61:60:c0)
> Internet Protocol Version 4, Src: 18.64.18.8, Dst: 192.168.100.102
▼ Transmission Control Protocol, Src Port: 443, Dst Port: 56922, Seq: 0, Ack: 1, Len: 0
Source Port: 443
Destination Port: 56922
[Stream index: 8]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3682223361
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2919762555
1000 = Header Length: 32 bytes (8)
Flags: 0x012 (SYN, ACK)
Window: 65535
[Calculated window size: 65535]

0000 f8 34 41 61 60 c0 f4 a4 d6 16 fc 6f 08 00 45 00 .4Aa*... ..o..E:
0010 00 34 00 00 40 00 f7 06 3a 6d 12 40 12 08 c0 a8 .4..@...:m@....
0020 64 66 01 bb de 5a db 7a 4c d1 ae 08 0a 7b 80 12 df...Z-z L....{..
0030 ff ff 64 f3 00 00 02 04 05 84 01 01 04 02 01 03 ..d.....
0040 03 09 ..

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ip.addr == 159.89.140.122

No.	Time	Source	Destination	Protocol	Length	Info
777	19.359826600	192.168.100.102	159.89.140.122	TCP	66	56936 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
778	19.661805900	159.89.140.122	192.168.100.102	TCP	66	443 → 56936 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
779	19.662000100	192.168.100.102	159.89.140.122	TCP	54	56936 → 443 [ACK] Seq=1 Ack=1 Win=66304 Len=0
781	19.663345500	192.168.100.102	159.89.140.122	TLSv1.3	571	Client Hello (SNI=mp0eff94b33b385b71a6.free.beeceptor.com)
783	19.864391700	159.89.140.122	192.168.100.102	TCP	54	443 → 56936 [ACK] Seq=1 Ack=518 Win=64128 Len=0
784	19.869054000	159.89.140.122	192.168.100.102	TLSv1.3	2878	Server Hello, Change Cipher Spec, Application Data
785	19.869191700	192.168.100.102	159.89.140.122	TCP	54	56936 → 443 [ACK] Seq=518 Ack=2825 Win=66304 Len=0
786	19.872808800	159.89.140.122	192.168.100.102	TLSv1.3	1161	Application Data, Application Data, Application Data, Application Data
787	19.914480900	192.168.100.102	159.89.140.122	TCP	54	56936 → 443 [ACK] Seq=518 Ack=3932 Win=65024 Len=0
788	19.917106700	192.168.100.102	159.89.140.122	TLSv1.3	118	Change Cipher Spec, Application Data
789	19.917437500	192.168.100.102	159.89.140.122	TLSv1.3	240	Application Data
792	20.173335300	159.89.140.122	192.168.100.102	TLSv1.3	109	Application Data
793	20.173338800	159.89.140.122	192.168.100.102	TLSv1.3	89	Application Data
794	20.173340800	159.89.140.122	192.168.100.102	TLSv1.3	85	Application Data

▼ Transmission Control Protocol, Src Port: 443, Dst Port: 56936, Seq: 0, Ack: 1, Len: 0
Source Port: 443
Destination Port: 56936
[Stream index: 29]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 652720568
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2993825867
1000 = Header Length: 32 bytes (8)
Flags: 0x012 (SYN, ACK)
Window: 64240
[Calculated window size: 64240]
Checksum: 0x87dc [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0

0000 f8 34 41 61 60 c0 f4 a4 d6 16 fc 6f 08 00 45 00 .4Aa*... ..o..E:
0010 00 34 00 00 40 00 32 06 f7 e1 9f 59 8c 7a c0 a8 .4..@...:m@....
0020 64 66 01 bb de 68 26 e7 b9 b8 b2 72 28 4b 80 12 df...h8...r(K..
0030 fa f0 87 dc 00 00 02 04 05 84 01 01 04 02 01 03 ..
0040 03 07 ..

Disini juga bisa dilihat bahwasannya datanya terlihat/ter tracking apabila di filter sesuai dengan karakteristik datanya.

*Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns.qry.name contains "beeceptor"

No.	Time	Source	Destination	Protocol	Length	Info
771	18.939275700	2001:448a:2083:6c40:b9...	2001:4489:208:102::2	DNS	119	Standard query 0x05d7 A mp0eff94b33b385b71a6.free.beeceptor.com
772	18.939766500	2001:448a:2083:6c40:b9...	2001:4489:208:102::2	DNS	119	Standard query 0x9372 AAAA mp0eff94b33b385b71a6.free.beeceptor.com
773	18.977612500	2001:448a:2083:6c40:b9...	2001:4489:202:102::2	DNS	119	Standard query 0x9372 AAAA mp0eff94b33b385b71a6.free.beeceptor.com
774	18.977942300	2001:448a:2083:6c40:b9...	2001:4489:202:102::2	DNS	119	Standard query 0x05d7 A mp0eff94b33b385b71a6.free.beeceptor.com
775	19.353517600	2001:4489:208:102::2	2001:448a:2083:6c40:b9...	DNS	183	Standard query response 0x9372 AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com
776	19.353520400	2001:4489:208:102::2	2001:448a:2083:6c40:b9...	DNS	135	Standard query response 0x05d7 A mp0eff94b33b385b71a6.free.beeceptor.com A 159.89.140.122
828	20.893415000	2001:4489:202:102::2	2001:448a:2083:6c40:b9...	DNS	135	Standard query response 0x05d7 A mp0eff94b33b385b71a6.free.beeceptor.com A 159.89.140.122
850	21.658794300	2001:4489:202:102::2	2001:448a:2083:6c40:b9...	DNS	183	Standard query response 0x9372 AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com
2033	44.406557600	2001:448a:2083:6c40:b9...	2001:4489:208:102::2	DNS	119	Standard query 0xa100 A mpcf8813a9c58c06c06b.free.beeceptor.com
2034	44.407001500	2001:448a:2083:6c40:b9...	2001:4489:208:102::2	DNS	119	Standard query 0xe8be AAAA mpcf8813a9c58c06c06b.free.beeceptor.com
2035	44.441588100	2001:448a:2083:6c40:b9...	2001:4489:202:102::2	DNS	119	Standard query 0xa100 A mpcf8813a9c58c06c06b.free.beeceptor.com
2036	44.441591200	2001:448a:2083:6c40:b9...	2001:4489:202:102::2	DNS	119	Standard query 0xe8be AAAA mpcf8813a9c58c06c06b.free.beeceptor.com
2041	44.850496700	2001:4489:208:102::2	2001:448a:2083:6c40:b9...	DNS	135	Standard query response 0xa100 A mpcf8813a9c58c06c06b.free.beeceptor.com A 159.89.140.122
2042	44.850499300	2001:4489:208:102::2	2001:448a:2083:6c40:b9...	DNS	183	Standard query response 0xe8be AAAA mpcf8813a9c58c06c06b.free.beeceptor.com SOA ns1.digitalocean.com

> Frame 776: Packet, 135 bytes on wire (1080 bits), 135 bytes captured (1080 bits) on interface \Device\NPF...
> Ethernet II, Src: HuaweiTechno_16:fc:6f (f4:a4:d6:16:fc:6f), Dst: Intel_61:60:c0 (f8:34:41:61:60:c0)
> Internet Protocol Version 6, Src: 2001:4489:208:102::2, Dst: 2001:448a:2083:6c40:b9c::ca57:38c:cc56
> User Datagram Protocol, Src Port: 53, Dst Port: 49623
▼ Domain Name System (response)
Transaction ID: 0x05d7
> Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
> Queries
> Answers
[Request In: 771]
[Time: 414.244700 milliseconds]

0000 f8 34 41 61 60 c0 f4 a4 d6 16 fc 6f 86 dd 60 00 .4Aa*... ..o..:
0010 00 00 00 51 11 3d 20 01 44 89 02 08 01 02 00 00 ...Q.=.D.....
0020 00 00 00 00 00 02 20 01 44 8a 20 83 6c 40 b9 9cD..l@..
0030 ca 57 03 8c cc 56 00 35 c1 d7 00 51 0b a3 05 d7 W...V.5...Q..
0040 81 80 00 01 00 01 00 00 00 00 14 6d 70 30 65 66mp0eff
0050 66 39 34 02 33 33 62 33 38 35 62 37 31 61 36 04 f94b33b3 85b71a6-
0060 66 72 65 65 00 62 65 65 63 65 70 74 6f 72 03 63 free-bee ceptor-c
0070 6f 6d 00 00 01 00 01 c0 0c 00 01 00 01 00 00 07 om.....
0080 08 00 04 9f 59 8c 7aY-z

Pembahasan Source Code

Menggunakan library ElementTree dan Pandas untuk membaca file XML, mengekstrak field ID/NIM, Nama, Prodi, Status dan Angkatan untuk Data 1 dan 2 kemudian membuat DataFrame, lalu menyimpan hasil parsing ke format CSV yang terstruktur.

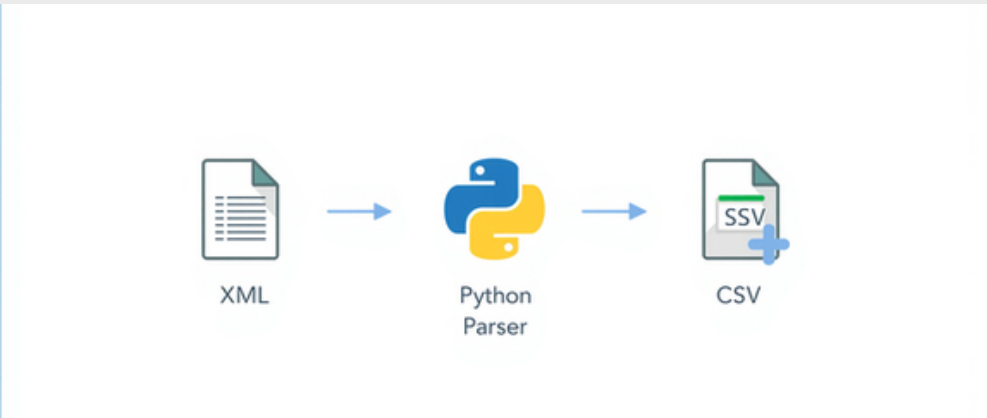
```
1  import xml.etree.ElementTree as ET
2  import pandas as pd
3
4  data = []
5
6  # FILE 1
7  tree = ET.parse("response-budi.xml")
8  root = tree.getroot()
9
10 user = root.find("user")
11
12 data.append({
13     "ID/NIM": user.find("id").text,
14     "Nama": user.find("name").text,
15     "Prodi": user.find("prodi").text,
16     "Status": user.find("Status").text,
17     "Angkatan": user.find("Angkatan").text
18 })
19
20 # FILE 2
21 tree = ET.parse("response-andi.xml")
22 root = tree.getroot()
23
24 mhs = root.find("Mahasiswa")
25
26 data.append({
27     "ID/NIM": mhs.find("NIM").text,
28     "Nama": mhs.find("Nama").text,
29     "Prodi": mhs.find("Prodi").text,
30     "Status": mhs.find("Status").text,
31     "Angkatan": mhs.find("Angkatan").text
32 })
33
34 df = pd.DataFrame(data)
35
36 print(df)
37
38 df.to_csv("hasil_parsing.csv", index=False)
```


Hasil Parsing CSV

Tabel hasil parsing menampilkan beberapa kolom bahwa pipeline XML → CSV berfungsi dengan baik.

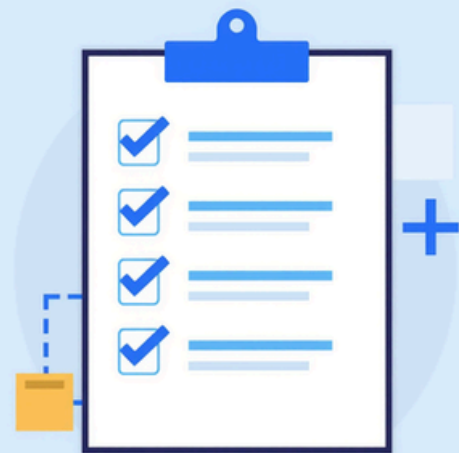
Hasil konversi data XML menjadi format CSV yang terstruktur menggunakan Python ElementTree dan Pandas.

Data XML berhasil dikonversi menjadi tabel terstruktur yang mudah dianalisis. Proses parsing berjalan sesuai dengan yang diharapkan.



hasil_parsing - Excel													
File Home WPS PDF Insert Page Layout Formulas Data Review View Help													
Clipboard Font Alignment Number Styles													
A1	ID/NIM												
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	ID/NIM	Nama	Prodi	Status	Angkatan								
2	10293	Budi Santoso	Informatika	Aktif	2025								
3	23010101	Andi Wijaya	Sains Data	Aktif	2025								
4													
5													
6													
7													
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9													
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14													
15													
16													
17													
18													
19													

	ID/NIM	Nama	Prodi	Status	Angkatan
0	10293	Budi Santoso	Informatika	Aktif	2025
1	23010101	Andi Wijaya	Sains Data	Aktif	2025



Hasil dan Evaluasi



HTTPS GET & POST

Metode GET dan POST berfungsi dengan benar, status 200 OK



Wireshark & XML Parsing

Paket jaringan berhasil dicapture, XML berhasil di-parse



Output CSV & Workflow

Data terkonversi ke CSV, workflow komunikasi sesuai tujuan proyek

Troubleshoot

- Kendala Filter Wireshark: Tidak bisa memfilter menggunakan keyword http secara langsung karena endpoint Beeceptor menggunakan protokol aman HTTPS/TLSv1.3
- Solusi: Menerapkan display filter spesifik menggunakan IP target (ip.addr == 159.89.140.122).
- Kendala Parsing Data (NaN): Muncul nilai missing value atau NaN pada dataset karena adanya perbedaan struktur payload antara data Andi dan data Budi.
- Solusi: Melakukan setting ulang dan penyesuaian pada kedua payload tersebut agar format dan strukturnya sama persis.
- Kendala Response Status Code: Endpoint POST awalnya mengembalikan status 200 OK, yang mana kurang tepat untuk aksi pembuatan data baru.
- Solusi: Mengubah arsitektur respon API menjadi status 201 Created agar sesuai dengan standar protokol HTTP yang semestinya.

Kesimpulan

Implementasi Berhasil



Implementasi communication protocol menggunakan HTTPS, Wireshark, XML, dan Python berjalan sukses sesuai tujuan proyek.

Tujuan Tercapai



Komunikasi data, analisis jaringan, dan transformasi data XML ke CSV tercapai secara efektif dan tujuan pembelajaran terpenuhi.

Kelompok 05

Program Studi Sains
Data

Aryadi, Faza, Uswatun, Nabel

Terima Kasih

Questions & Discussion
Kelompok 05 – UTS
Communication Protocol

